

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 1 (Generated practice paper - Set 30)**

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

0 1

In 2020 the tanker MV Wakashio ran aground on a coral reef off Mauritius and leaked about 1000 tonnes of fuel oil into a shallow lagoon near mangroves and a marine park.

0 1

. 1

Explain why inflatable booms were deployed immediately, and give **one** limitation of booms.

[3 marks]

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0 1

. 2

Describe how skimmers and absorbent materials remove oil, and state how the collected oil is disposed of.

[3 marks]

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0 1

. 3

Explain why dispersants were **not** used in the lagoon.

[2 marks]

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0 1

. 4

Explain how bioremediation could help remove the oil that remained on the mangrove sediments, and suggest how it could be accelerated.

[2 marks]

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10

0 2

Table 1 shows the volume of water in some reservoirs of the hydrological cycle and the annual transfer rates through them.

Table 1

Reservoir	Volume / $\times 10^{15}$ dm^3	Annual transfer (inflow = outflow) / $\times 10^{15} \text{ dm}^3 \text{ yr}^{-1}$	Percentage of global water / %
Oceans	1 340 000	425	97
Ice caps and glaciers	24 000	3	1.7
Groundwater	23 000	12	1.7
Lakes and rivers	180	40	0.013
Soil moisture	17	80	0.001
Atmosphere	13	480	0.001

0 2 . 1

The residence time of water in a reservoir is given by: residence time = volume in reservoir / annual transfer rate. Use Table 1 to calculate the residence time of water in the atmosphere. Give your answer to the nearest whole day.

[3 marks]

Answer

days

0 2 . 2

Calculate the residence time of water in the oceans.

[1 mark]

Answer years

0 2 . 3

Explain why the atmosphere is an important reservoir even though it holds only 0.001% of global water.

[2 marks]

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0 2 . 4

Explain why human changes to a small reservoir such as soil moisture are more obvious than changes to the oceans.

[2 marks]

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0 2 . 5

Explain why the hydrological cycle is described as being in dynamic equilibrium.

[2 marks]

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10

03 . 1

Describe **three** factors that have restricted the growth of nuclear power.

[3 marks]

1

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2

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3

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03 . 2

Explain how a plutonium 'breeder' reactor can produce more fissile fuel than it uses.

[3 marks]

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03 . 3

Give **two** advantages and **two** disadvantages of thorium reactors compared with uranium reactors.

[4 marks]

Advantages

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Disadvantages

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10

0 4 . 1

Explain what is meant by the **embodied energy** of a vehicle and how it can be reduced.

[3 marks]

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0 4 . 2

Describe **four** principles of designing vehicles for end of life.

[4 marks]

1

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2

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3

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4

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0 4 . 3

Explain why there is an optimum speed for fuel efficiency in a car.

[2 marks]

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0 4 . 4

Give **one** way driver behaviour or vehicle systems can reduce fuel use in congested traffic.

[1 mark]

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10

0 5 . 1

Explain how each of the following human activities changes the amount of carbon in the atmosphere.

[6 marks]

Ploughing farmland

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Draining marshland / peat bogs

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Toxic pollution of the sea

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0 5 . 2

Give **three** anaerobic environments created by human activities that release methane.

[3 marks]

1

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2

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3

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0 5 . 3

Explain how rising atmospheric CO₂ causes ocean acidification.

[1 mark]

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10

06 . 1

A student measured the pH of a stream with a meter that reads to 0.01 pH units but had not been calibrated. Which statement is correct? Tick **one** box.

[1 mark]

A The readings are precise but may not be accurate

☐

B The readings are accurate but not precise

☐

C The readings are both accurate and precise

☐

D The readings are reliable because the meter is electronic

☐

06 . 2

Explain what is meant by an **anomalous result** and how a student should decide whether to include it.

[2 marks]

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06 . 3

Explain why a preliminary study is carried out before the main investigation.

[2 marks]

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5

07 . 1

Explain how each of the following human activities increases soil erosion.

[4 marks]

Overgrazing

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Reducing soil biota

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07 . 2

Explain why landslides become more likely on steep slopes some years after deforestation.

[2 marks]

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07 . 3

Describe **four** environmental impacts of accelerated soil erosion away from the field where it occurs.

[4 marks]

1

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2

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3

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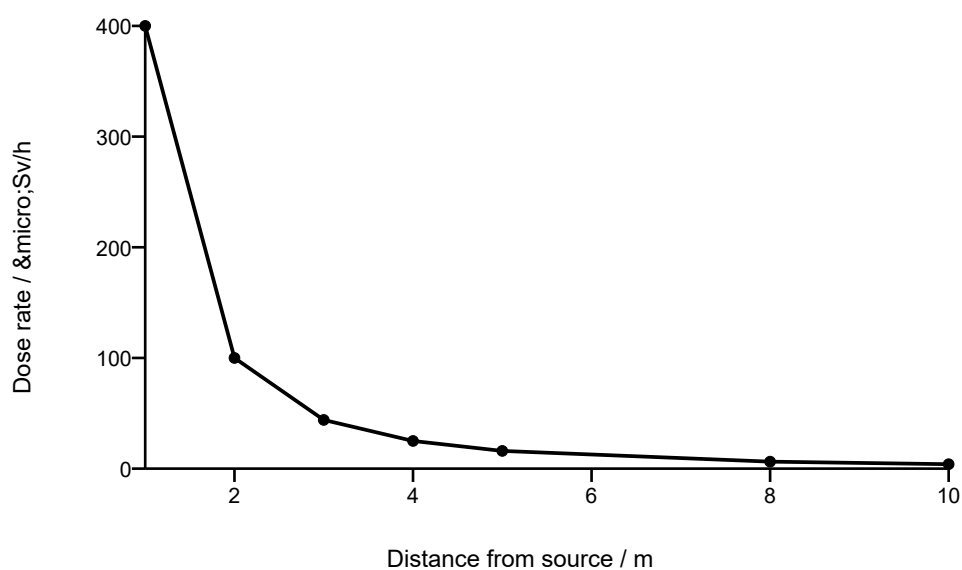
4

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10

0 8

Figure 1 shows the radiation dose rate measured at different distances from a small gamma source.

Figure 1**0 8****1**

Use Figure 1 to show that the dose rate follows the inverse square law.

[2 marks]**0 8****2**

Calculate the dose rate at 20 m from the source.

[1 mark]**Answer**

 $\mu\text{Sv/h}$ **0 8****3**

Explain how each of the following reduces the exposure of workers handling radioactive sources.

[4 marks]

Closed (sealed) sources

Absorbing materials

Reduced exposure time

Worker monitoring

08 . 4

Explain why potassium iodide tablets are issued to people near a nuclear accident.

[2 marks]

08 . 5

State the meaning of ALARA.

[1 mark]

10

09

Figure 1 is a flow diagram of the main stages in a sewage treatment works using activated sludge treatment.

Figure 1

Stage	Process
Pre-treatment	Screens and grit traps
Primary treatment	Primary sedimentation
Secondary treatment	Aeration tank then secondary sedimentation
Tertiary treatment	Phosphate stripping / micro-strainers / UV
Sludge treatment	Anaerobic digestion

09 . 1

Explain the purpose of screens and grit traps.

[2 marks]

09 . 2

Explain why primary sedimentation is described as the most important process.

[2 marks]

09 . 3

Describe what happens in the aeration tank and explain why the sludge from the secondary sedimentation tank is returned to it.

[3 marks]

09 . 4

Explain why tertiary treatment may be needed before effluent is discharged, and describe **one** tertiary process.

[3 marks]

10

1	0	.	1
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Evaluate the success of the methods and strategies used to reduce global climate change.

[9 marks]

Extra space

10 . 2 State what is meant by a **carbon footprint**.

[1 mark]

10

Write an essay on one of the following topics.

$$\boxed{1} \boxed{1} . \boxed{1}$$

Discuss how knowledge of the properties of pollutants can be used to reduce the severity of their impact on the environment.

[25 marks]

OR

1 1 . 2

Soil is often described as a non-renewable resource. Discuss the causes and consequences of soil degradation and evaluate the strategies that can be used to manage soils sustainably.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

Do not write outside the box

A large rectangular box containing 30 horizontal dotted lines for writing.

[illegible]

END OF QUESTIONS